Deployment of a Web Server on Amazon Web Services (AWS) EC2

Task 2 – Cloud Computing Practical Report

Submitted by: Habiba Shafait

Alfido Tech

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1. Objective

The objective of this task is to launch a virtual server (EC2 instance) on Amazon Web Services (AWS), install and configure the Apache2 web server on it, and verify successful deployment by accessing the server's default web page through a public IP address. This exercise demonstrates practical skills in cloud infrastructure provisioning, remote server administration via SSH, and basic web server configuration.

2. Tools and Environment Used

- Amazon Web Services (AWS) Management Console – EC2 Service
- Amazon Machine Image (AMI): Ubuntu Server 24.04 LTS
- Instance Type: t3.micro (Free Tier eligible)
- Region: US East (N. Virginia)
- Web Server Software: Apache2
- Access Method: SSH terminal (EC2 Instance Connect) and Web Browser

3. Step-by-Step Procedure

Step 1: Accessing the AWS EC2 Console

The AWS Management Console was accessed and the EC2 service dashboard was opened. This service provides the interface for launching, managing, and monitoring virtual server instances (Amazon EC2) in the cloud.

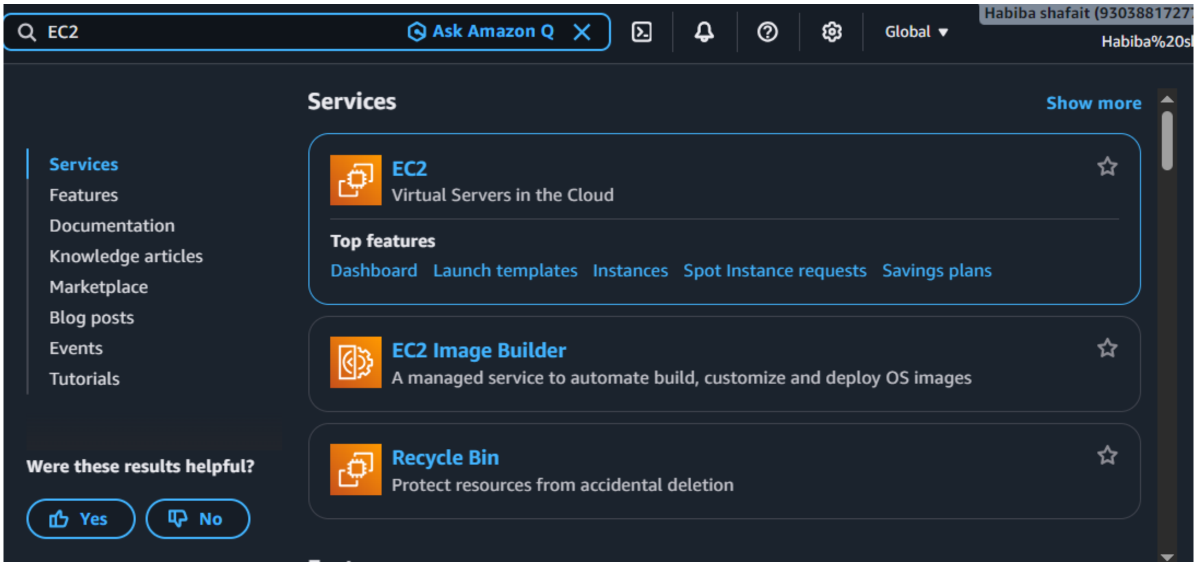

Figure 1: AWS EC2 service dashboard

Step 2: Reviewing EC2 Resources

Before launching any instance, the EC2 Resources panel was checked in the N. Virginia region. Initially, no running instances, volumes, or other resources existed under this account, confirming a clean starting environment.

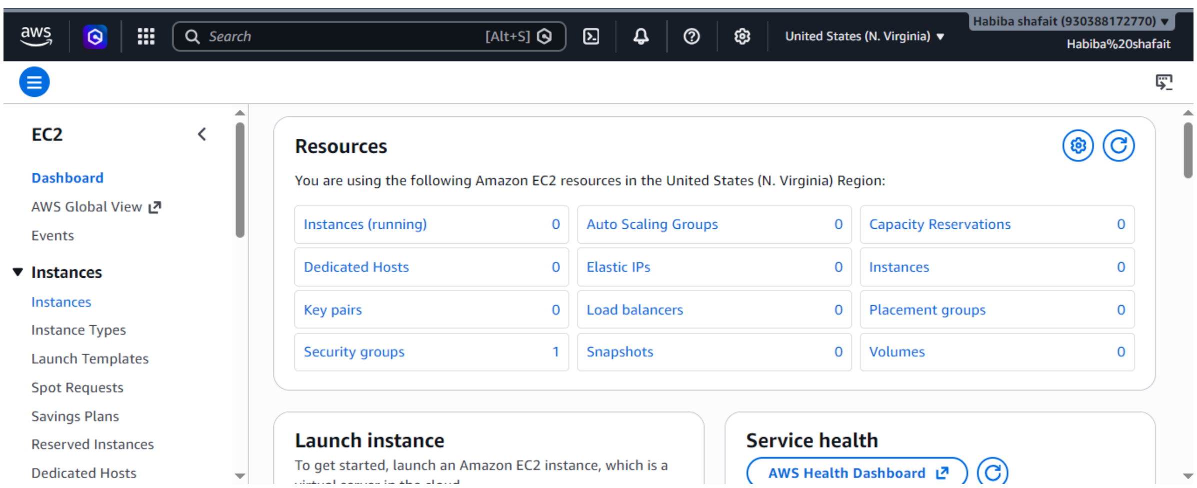

Figure 2: EC2 resource overview showing zero active resources

Step 3: Confirming No Existing Instances

The Instances panel was opened to verify that no EC2 instance was already running, ensuring the new instance would be created from scratch.

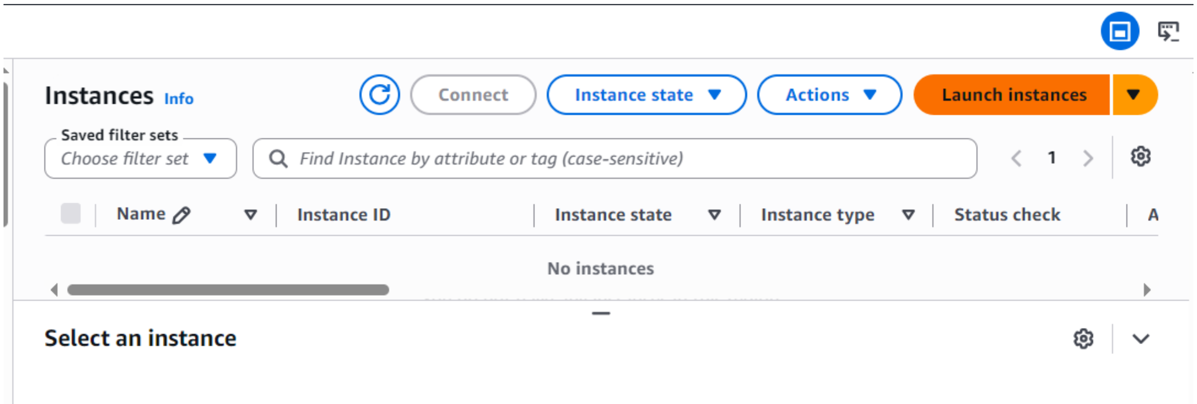

Figure 3: Instances panel showing no existing instances

Step 4: Launching a New Instance and Selecting the AMI

A new instance was launched using the 'Launch Instance' wizard. The Ubuntu Server 24.04 LTS (HVM) Amazon Machine Image (AMI) was selected, and the instance type was set to t3.micro, which is Free Tier eligible.

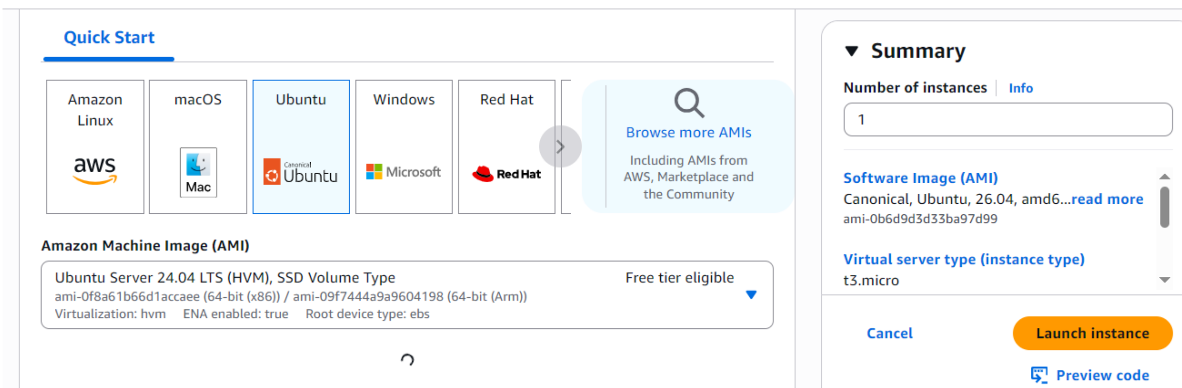

Figure 4: AMI selection – Ubuntu Server 24.04 LTS with t3.micro instance type

Step 5: Reviewing Instance Type Specifications

The specifications of the t3.micro instance type were reviewed, including 2 vCPUs, 1 GiB memory, and on-demand pricing details for various operating systems, confirming it was suitable and free-tier eligible for this deployment.

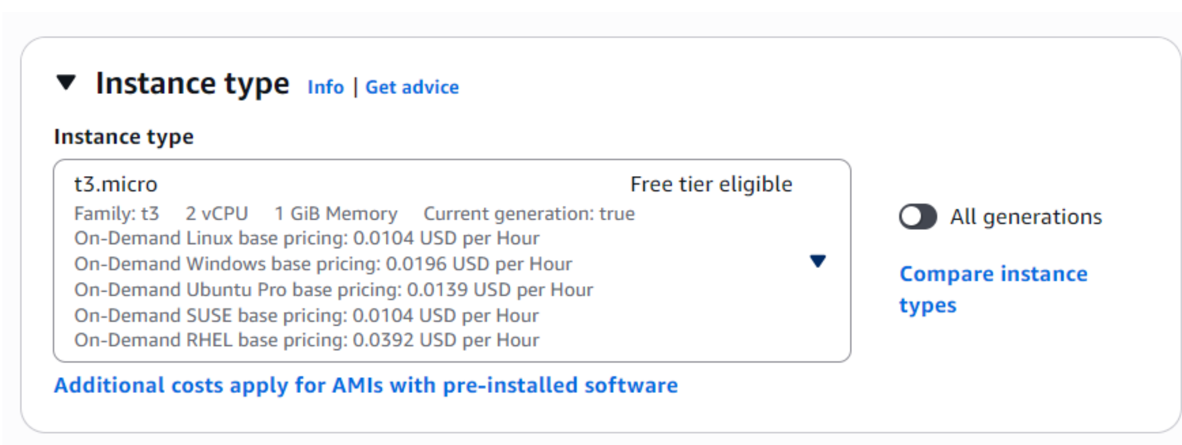

Figure 5: t3.micro instance type details and pricing

Step 6: Configuring the Key Pair

A key pair (MyWebServerKey) was selected for secure SSH access to the instance. The key pair ensures that only authorized users can connect to the server remotely.

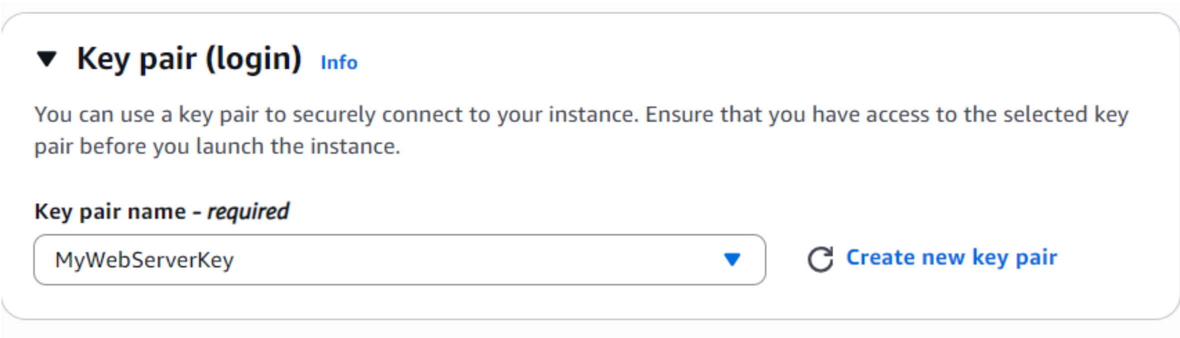

Figure 6: Key pair selection for secure instance login

Step 7: Verifying Successful Instance Launch

After launching, the instance named 'MyWebServer' appeared in the Instances panel with a 'Running' state, instance type t3.micro, and 3/3 status checks passed — confirming the instance was healthy and operational.

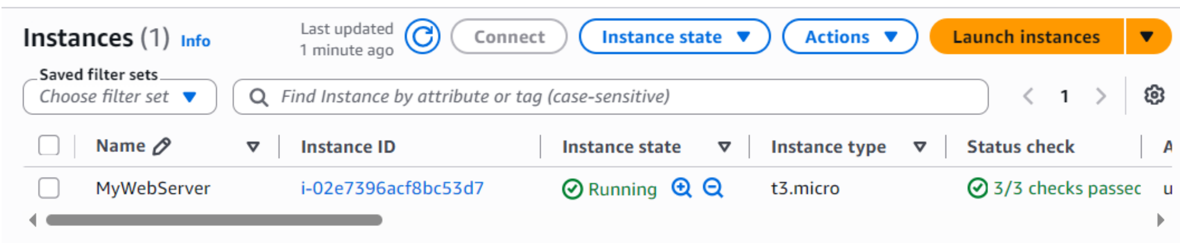

Figure 7: MyWebServer instance running with all status checks passed

Step 8: Connecting to the Instance

The 'Connect to instance' option was used to establish an SSH session. The instance's public IPv4 address (50.17.125.153) was noted, and the default username 'ubuntu' (standard for Ubuntu AMIs) was used to authenticate.

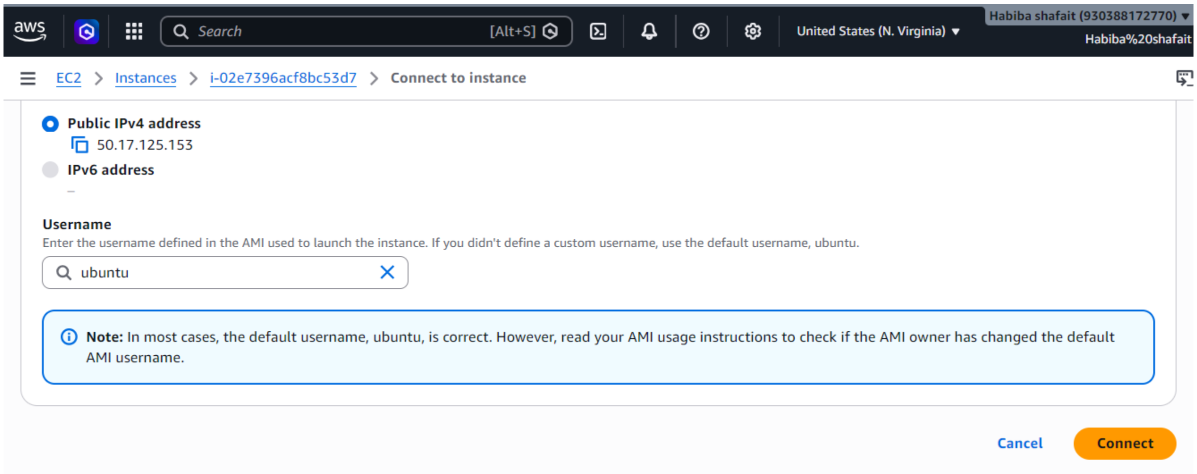

Figure 8: Connect-to-instance screen showing public IPv4 address and username

Step 9: Updating Package Repositories

Once connected via the terminal, the command 'sudo apt update' was executed to refresh the local package index and ensure the latest package information was available before installing new software.

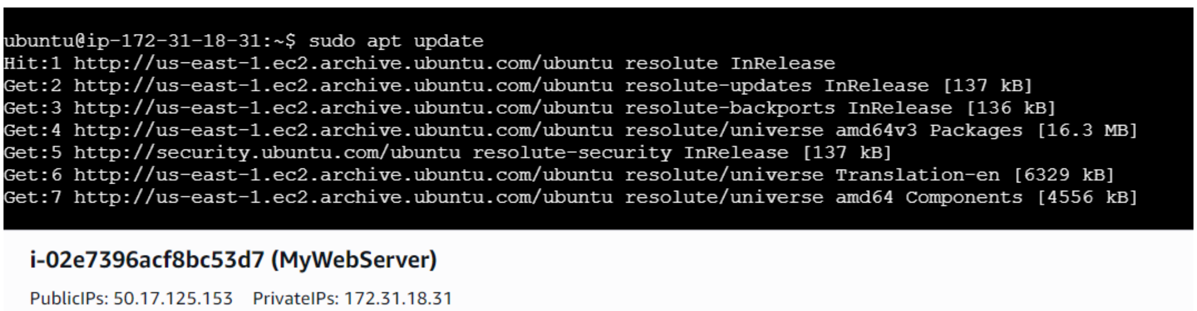

Figure 9: Terminal output of 'sudo apt update'

Step 10: Installing the Apache2 Web Server

The Apache2 web server package was installed using the command 'sudo apt install apache2 -y', which automatically installed Apache2 along with its required dependencies.

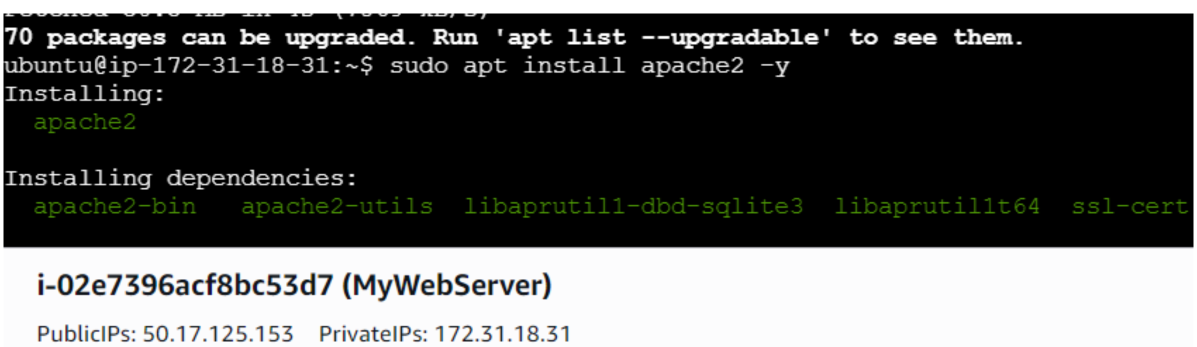

Figure 10: Installing Apache2 via apt

Step 11: Verifying the Apache2 Service Status

The command 'sudo systemctl status apache2' was run to confirm that the Apache2 service had started successfully. The output showed the service as 'active (running)', confirming the web server was operational on the instance.

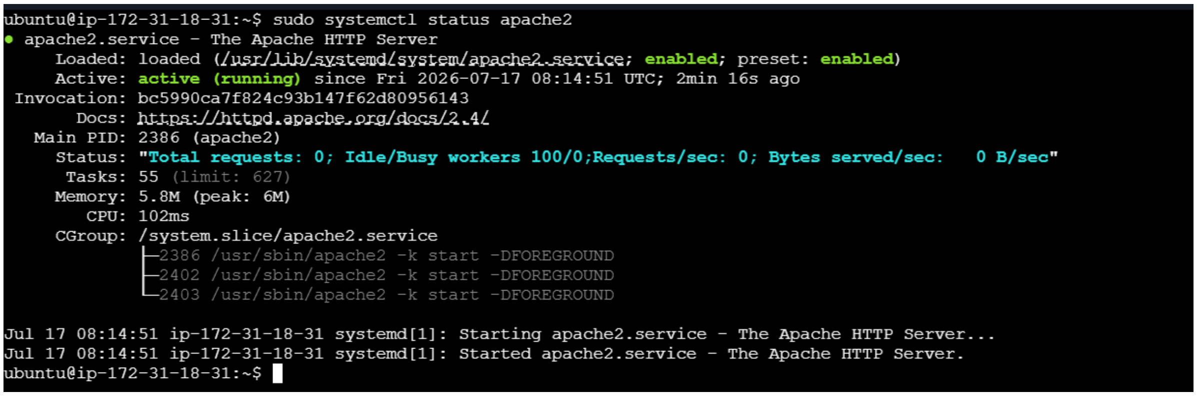

Figure 11: Apache2 service status confirming it is active and running

Step 12: Reviewing Instance Details

The instance summary page was reviewed, confirming the running state of 'MyWebServer' along with its public IPv4 address (50.17.125.153) and private IPv4 address (172.31.18.31).

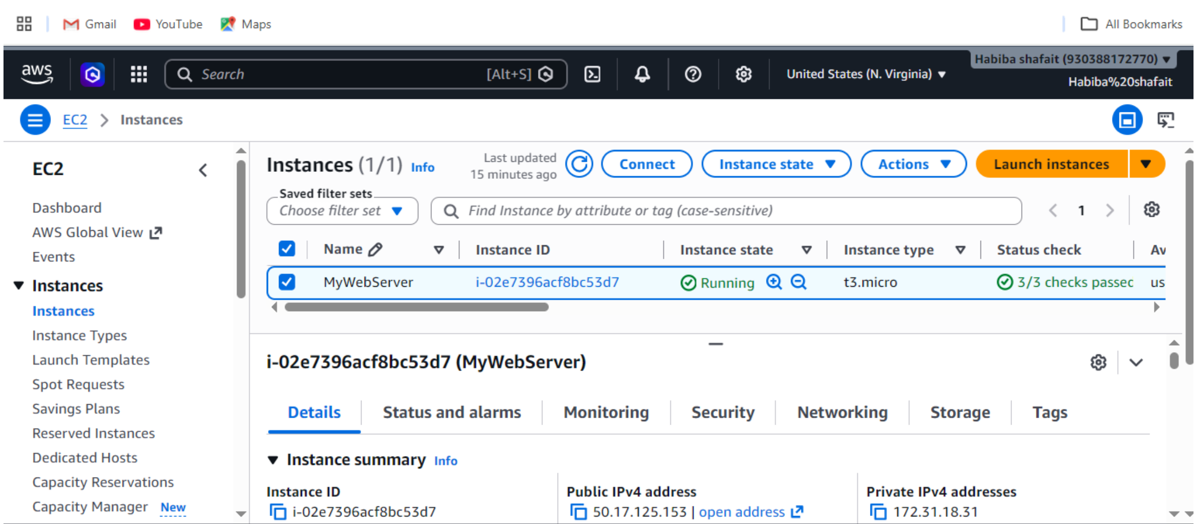

Figure 12: Instance summary showing public and private IP addresses

Step 13: Accessing the Web Server via Browser (Initial Check)

The public IP address of the instance was entered into a web browser. The browser tab title 'Apache2 Ubuntu Default Page: It works' confirmed that the HTTP request reached the server successfully.

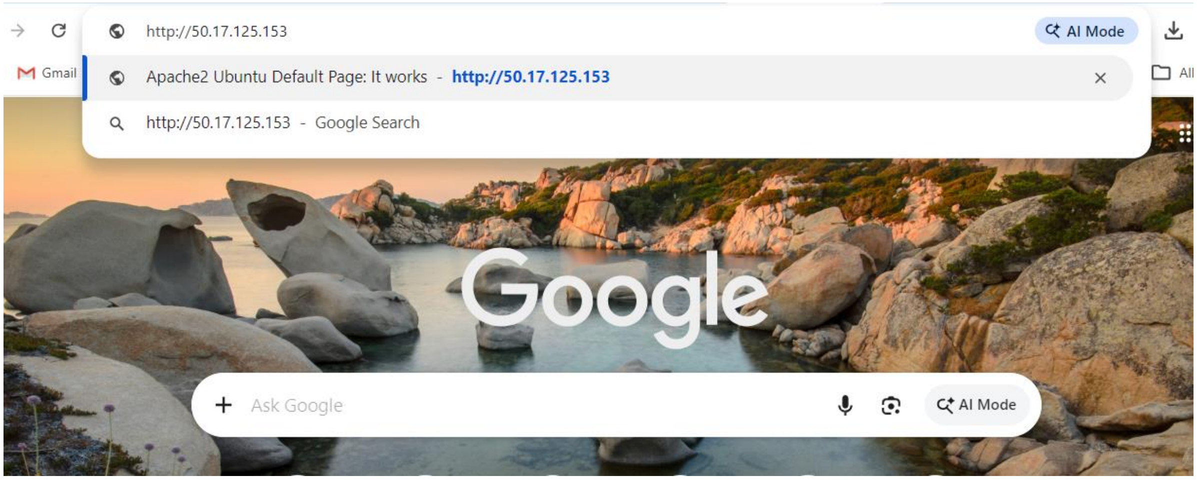

Figure 13: Browser tab confirming the Apache2 default page loaded

Step 14: Verifying the Apache2 Default Web Page

Finally, the full Apache2 Ubuntu default web page was loaded in the browser at http://50.17.125.153, displaying the standard welcome message. This confirmed that the web server was successfully deployed, running, and publicly accessible over the internet.

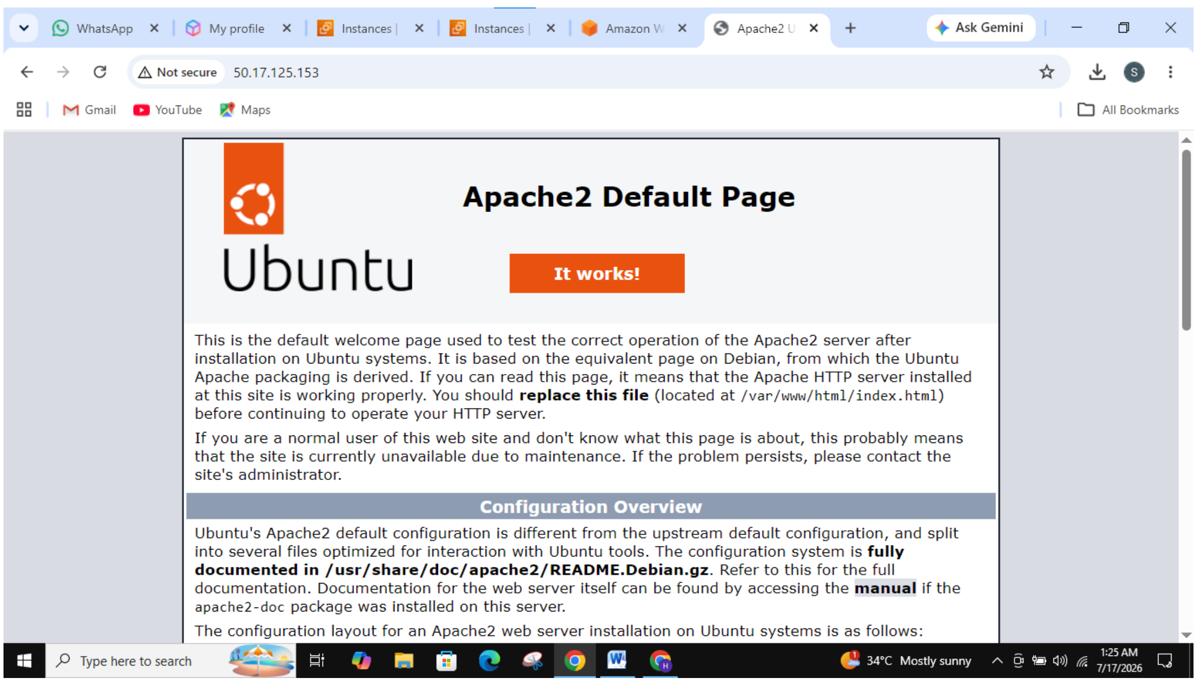

Figure 14: Apache2 default welcome page accessed via public IP address

4. Conclusion

In this task, an Ubuntu Server 24.04 LTS instance was successfully launched on AWS EC2 using the Free Tier t3.micro instance type. The Apache2 web server was installed and configured on the instance via SSH, and its status was verified to be active and running. Finally, the web server was confirmed to be publicly accessible by loading its default page through the instance's public IP address in a web browser. This exercise provided hands-on experience with core AWS EC2 operations, remote Linux server administration, and basic web server deployment — foundational skills for cloud computing and IT infrastructure roles.